
COMP 3059 – Capstone Project I**Software Requirements Analysis and Design Assignment**

This assignment is an overview to gather the software needs with requirements analysis and help to proceed with the design.

The requirements analysis helps to break down functional and non-functional requirements to a basic design view to provide a clear system development process framework. It involves various entities, including business, stakeholders and technology requirements.

The design is the activity following requirements specification and before programming. Software design usually involves problem solving and planning a software solution.

To work on this assignment, you could use the references and a sample template given below. The sample template can be customized to suit the nature of your project.

Reference Reading/Example to work on this Sprint document:

www.cse.msu.edu/~chengb/RE-491/Papers/SRSEExample-webapp.doc

Source for this template:

www.tricity.wsu.edu/~mckinnon/cpts322/cpts322-srs-v1.doc

1.0 Introduction

This document defines the software requirements analysis and design for the ABC Restaurant HCM System. It specifies the functional and non-functional features required for the solution, identifies constraints, and outlines the logical design necessary for implementation.

It serves as a technical foundation for development and verification, providing a clear framework to guide coding, testing, and deployment.

1.1 Purpose

The ABC Restaurant HCM System is designed for small-to-medium-sized restaurants that need an integrated platform to manage employee schedules, payroll, time-off requests, and expense reimbursements.

Unlike enterprise solutions such as Dayforce HCM, this system offers a low-cost, user-friendly web application that streamlines routine management tasks while ensuring accuracy and compliance.

1.2 Scope

The ABC Restaurant HCM System is designed for small-to-medium-sized restaurants that need an integrated platform to manage employee schedules, payroll, time-off requests, and expense reimbursements.

Unlike enterprise solutions such as Dayforce HCM, this system offers a low-cost, user-friendly web application that streamlines routine management tasks while ensuring accuracy and compliance.

In Scope:

- Employee management (availability, clock-in/out, shift history)
- Schedule creation, update, and publishing
- Time-off and reimbursement workflows
- Payroll summary generation and export (PDF/Excel)
- Notifications and reminders
- Role-based authentication (Employee, Manager, Finance Staff, Admin)

Out of Scope:

- Integration with third-party payroll processors
- AI-based forecasting or analytics
- Mobile-only version (future phase)

2.0 System Overview

The system provides a centralized portal where restaurant managers, employees, and finance staff interact.

- **Managers** handle schedules, leave approvals, payroll reviews, and expense requests.
- **Employees** view shifts, submit requests, and clock in/out.
- **Finance Staff** receive payroll summaries and verify compliance.
- **Admins** maintain accounts and monitor system health.

The software will follow a **three-tier web architecture** (presentation, application, data) built for eventual deployment on a cloud or on-prem server.

2.1 Project Perspective

- This is a new self-contained system inspired by existing enterprise HCM solutions but re-engineered for smaller restaurant operations.
- It will initially run as a web-based prototype accessed through a browser, with future adaptability for Android or cross-platform clients.

2.2 System Context

- The application resides on a secure web server with database support.
- Users access it via standard browsers (Chrome, Edge, Firefox).
- External entities include restaurant employees, managers, and finance staff.
- Communication occurs through HTTP(S) requests and email notifications.

2.3 General Constraints

- Must operate within restaurant Wi-Fi or broadband environments.
- Must support 24 × 7 availability with minimal downtime.
- Backend must use an open-source stack (Spring Boot / PostgreSQL / Redis).
- All sensitive data (payroll, credentials) must be encrypted.
- Browser compatibility: Chrome ≥ v110, Edge ≥ v110.

2.4 Assumptions and Dependencies

- Restaurants possess reliable internet access.
- Managers and employees have basic digital literacy.
- The database server remains available and backed up daily.
- Payroll compliance logic will depend on provincial labor regulations (Ontario).
- Future integration with mobile apps will reuse the same REST API.

3.0 Functional Requirements

This section describes specific features of the software project. If desired, some requirements may be specified in the use-case format and listed in the Use Cases Section.

3.1.1 Request System

- Introduction
 - The Request System will consist of a Request Database, and two pages; The Requests page in the Employee view, and the Manage Requests page in the Manager view.
- Inputs
 - Employees can submit Requests, Managers can approve or deny Requests.
- Processing
 - When a Request is submitted, it is persisted to the Request Database, with all relevant details, and a default value of outstanding for the status.
 - When a Request has its status changed by a Manager, the Request Database is updated to reflect the change.
- Outputs
 - Whenever a Request is persisted to the Request Database, or its status has been updated, that is immediately reflected in the Manage Requests (Manager view) and Requests pages.

3.1.2 Schedule System

- Introduction
 - The Schedule System will consist of a Schedule Database and two pages, one for viewing in the Employee view and one for creating, managing and viewing in the Manager view.
- Inputs
 - Managers can create, edit, or delete Schedules.
 - Employees can view their schedules and submit availability.
- Processing
 - When a Manager creates or modifies a Schedule, the details are persisted to the Schedule Database, associating Employees with specific shifts and time slots.
 - When Employees submit availability, their data is stored and can be referenced by managers when creating or generating schedules.
- Outputs
 - Updated and created Schedules are immediately visible on the Schedules pages for both Employees and Managers.
 - Employees can receive real time updates if their shifts are changed.

3.1.2 Earnings/Payroll System

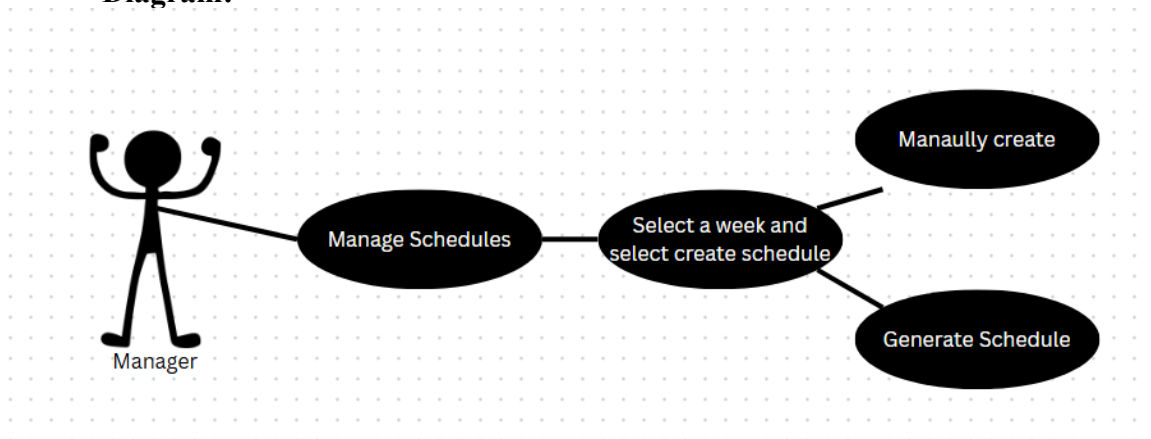
- Introduction

- The Earnings/Payroll System will consist of a Payroll Database and two pages, one for viewing earnings as an Employee and one for viewing and editing Employee earnings as a Manager.
- Inputs
 - Employee work hours are recorded from completed schedules.
 - Managers can make manual adjustments to pay rates, and hours accrued.
- Processing
 - The system calculates gross pay based on recorded work hours and employee pay rates.
- Outputs
 - Employees can view their calculated earnings, including weekly and yearly summaries through the Earnings page.
 - Managers can generate payroll reports and download data for accounting and payment.
 - Any updates to pay or recorded hours are immediately reflected in both the database and the Employee view.

3.2 Use Cases

3.2.1 Use Case #1 Create Staff Schedule

Diagram:



Brief Description:

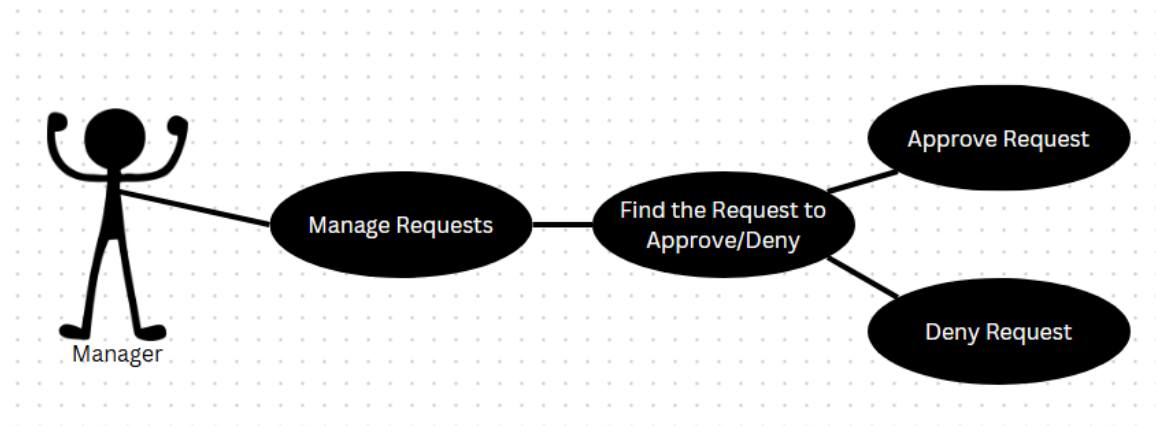
The Manager either manually creates or procedurally generates a weekly schedule based on availability submitted by the Employees.

Initial Step-By-Step Description:

1. The Manager selects Manage Schedules from their dashboard.
2. The Manager selects the week for which they would like to create a schedule.
3. The Manager selects Create Schedule.
4. If the Manager is generated a schedule, they will select generate schedule and it will be generated using the submitted availabilities, otherwise they will manually fill out each day in the week with employee shifts.

3.2.2 Use Case #2 Approve Or Deny Requests

Diagram:



Brief Description:

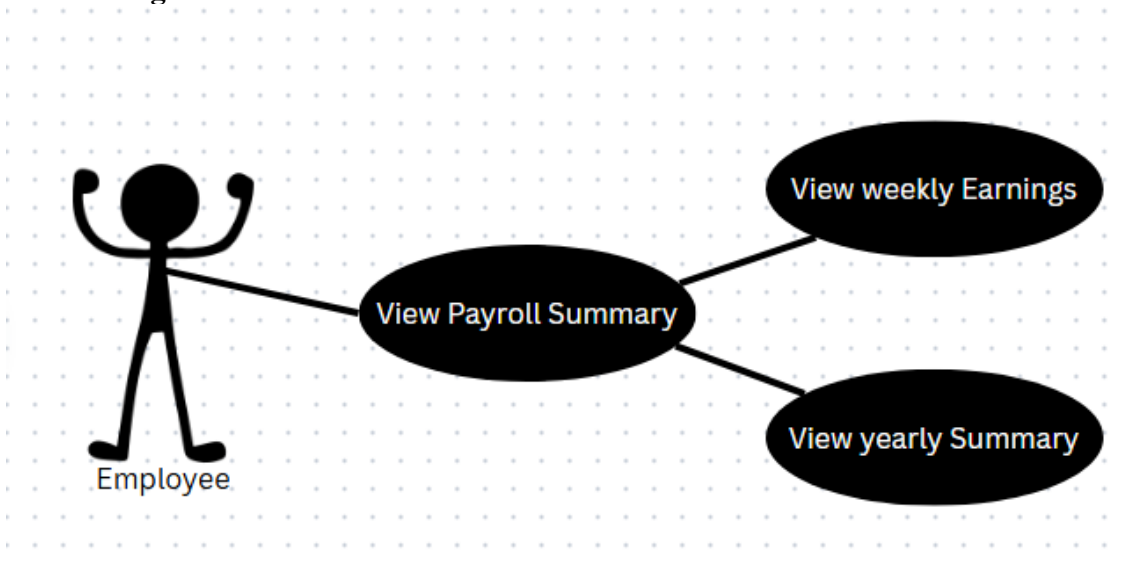
The Manager views and approves or denies various requests submitted by employees (time off, reimbursement, shift swap)

Initial Step-By-Step Description:

1. The manager selects Manage Requests
2. The Manager can select Approve or Deny from a list of outstanding Requests, as well as view their relevant details (Who it was submitted by, the type of Request, when it was submitted, it's current status)

3.2.3 Use Case #3 View Earnings

Diagram:



Brief Description:

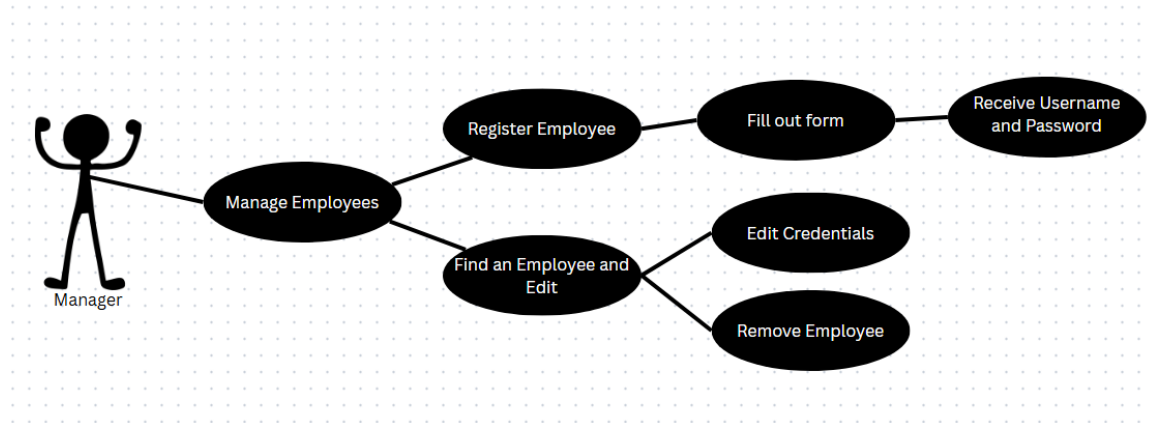
The employee can view their accrued hours and earnings, and view yearly summaries used for filling out tax documents

Initial Step-By-Step Description:

1. The Employee selects View payroll summary
2. The Employee either selects view weekly earnings, or view yearly earnings summary

3.2.4 Use Case #4 Manage User Accounts

Diagram:



Brief Description:

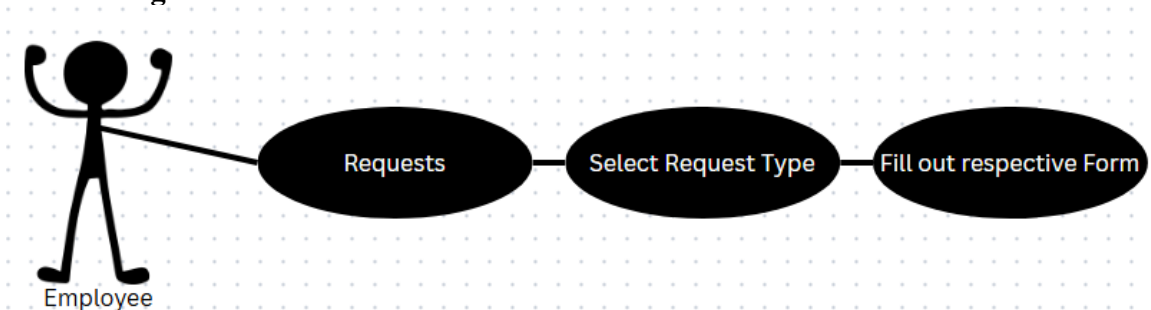
The Manager can view and manage the accounts of all Employees, as well as add new ones to the system.

Initial Step-By-Step Description:

1. The Manager selects Manage Employees from their dashboard.
2. If the Manager is:
 - a. Adding a new Employee Account.
 - i. The Manager selects Register Employee.
 - ii. The Manager fills out a form including Employee name, email, role.
 - iii. The Manager receives a generated username and password which is then given to the employee.
 - b. Managing an Employee Account.
 - i. The Manager selects Edit from one of the Employee Accounts listed on the screen.
 - ii. The Manager can then change the accounts credentials, or remove them from the system.

3.2.5 Use Case #5 Submit Requests

Diagram:

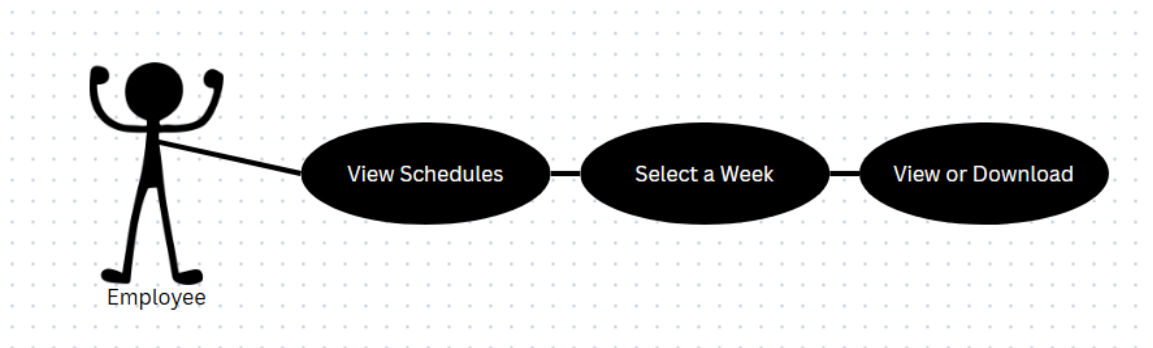


Brief Description:

Employees use this feature to submit various requests such as time-off, schedule change, or reimbursement. Managers are notified to review and approve or deny them.

Initial Step-By-Step Description:

1. The Employee logs in and navigates to “Requests.”
2. The Employee selects the type of request.
3. Time-Off Request (date range, reason)
4. Schedule Change Request (proposed swap or shift change)
5. Reimbursement Request (expense details, receipt upload)
6. The system displays the appropriate form based on selection.
7. The Employee fills in required fields and submits the form.
8. The system validates the information (e.g., ensures no overlapping shifts).
9. The request is saved to the database with status = Pending.
10. The Manager receives a notification of the new request.
11. Once reviewed, the Manager’s decision (Approved/Denied) is updated, and the Employee receives a notification.

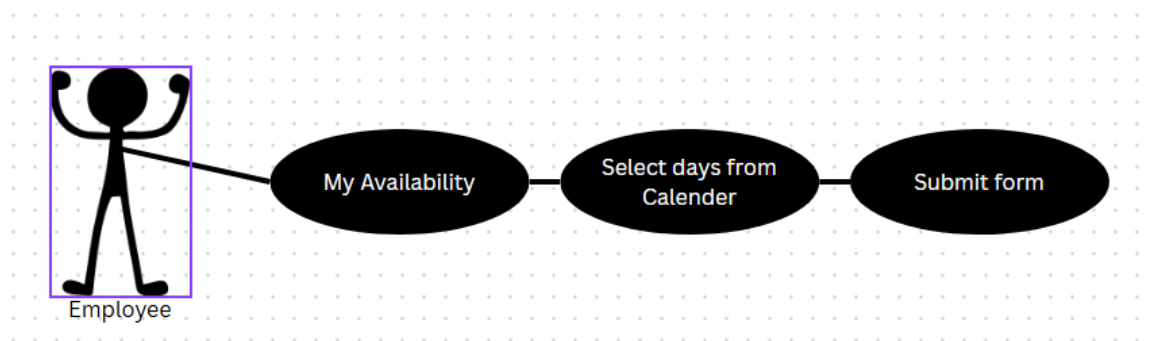
3.2.6 Use Case #6 View Schedules**Diagram:****Brief Description:**

Employees can view weekly schedules generated or created by the Manager.

Initial Step-By-Step Description:

1. The Employee selects View Schedules from their dashboard.
2. The Employee can then select a week from a list of weeks on their screen.
3. The Employee will then see a schedule for the given week which they can download as an image or pdf.

3.2.7 Use Case #7 Submit Availability**Diagram:**

**Brief Description:**

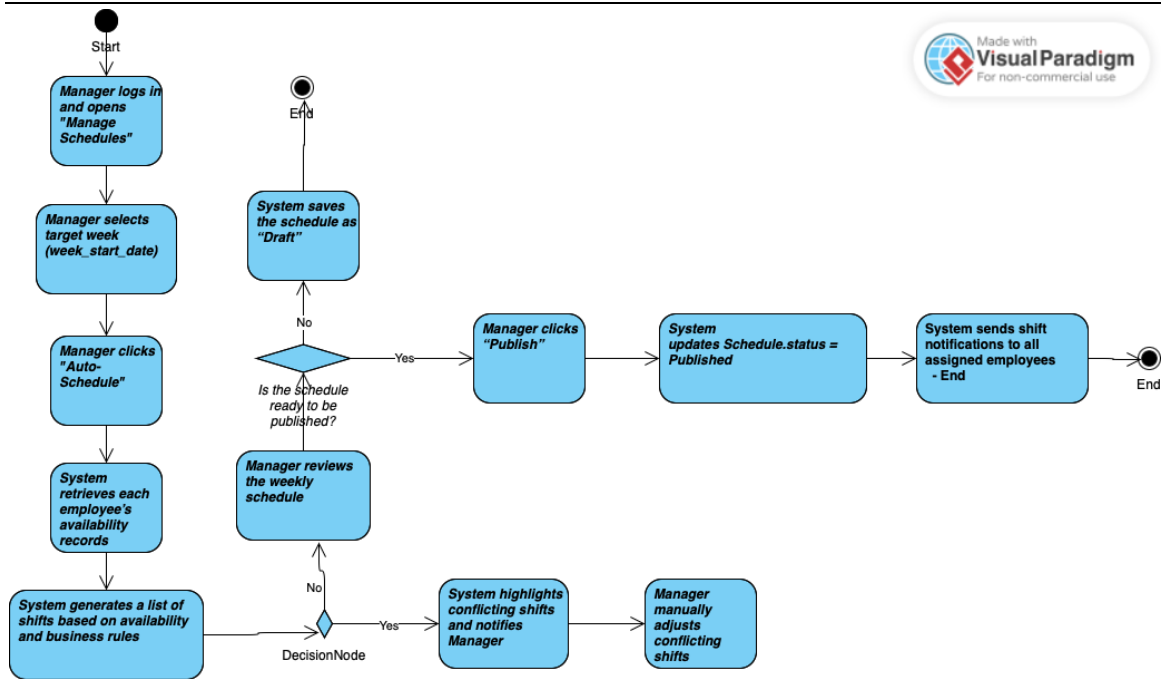
Employees can fill out a form containing

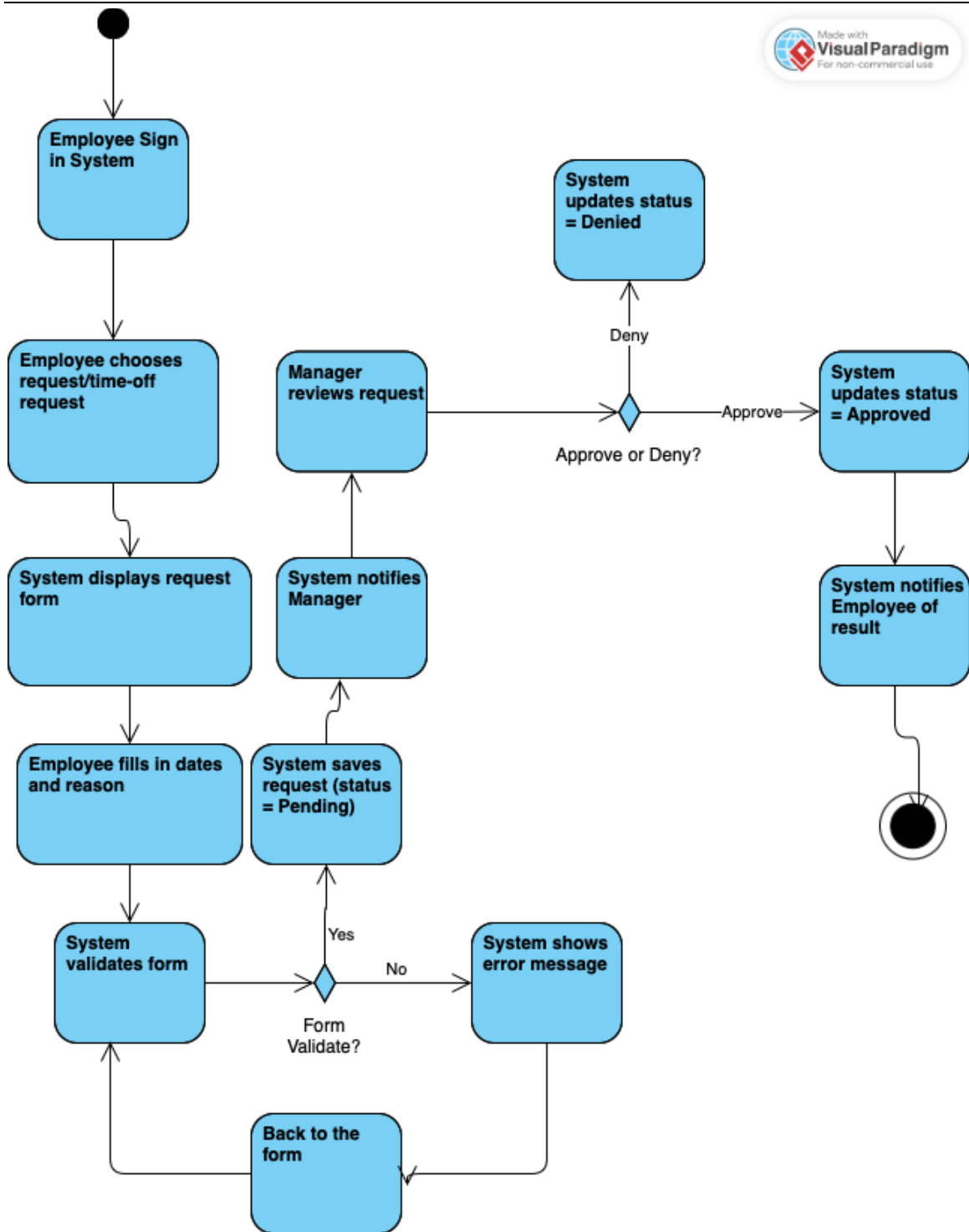
Initial Step-By-Step Description:

1. The Employee logs in and selects “My Availability.”
2. The system presents a weekly calendar view.
3. The Employee selects available days and time ranges (e.g., Mon 9–5, Wed 12–close).
4. The Employee submits the form.
5. The system validates entries to ensure logical ranges (no overlaps or invalid hours).
6. The system saves the updated availability record linked to the employee profile.
7. The Manager’s dashboard updates to reflect the latest availability for scheduling.
8. The system sends a confirmation message: “Your availability has been updated.”

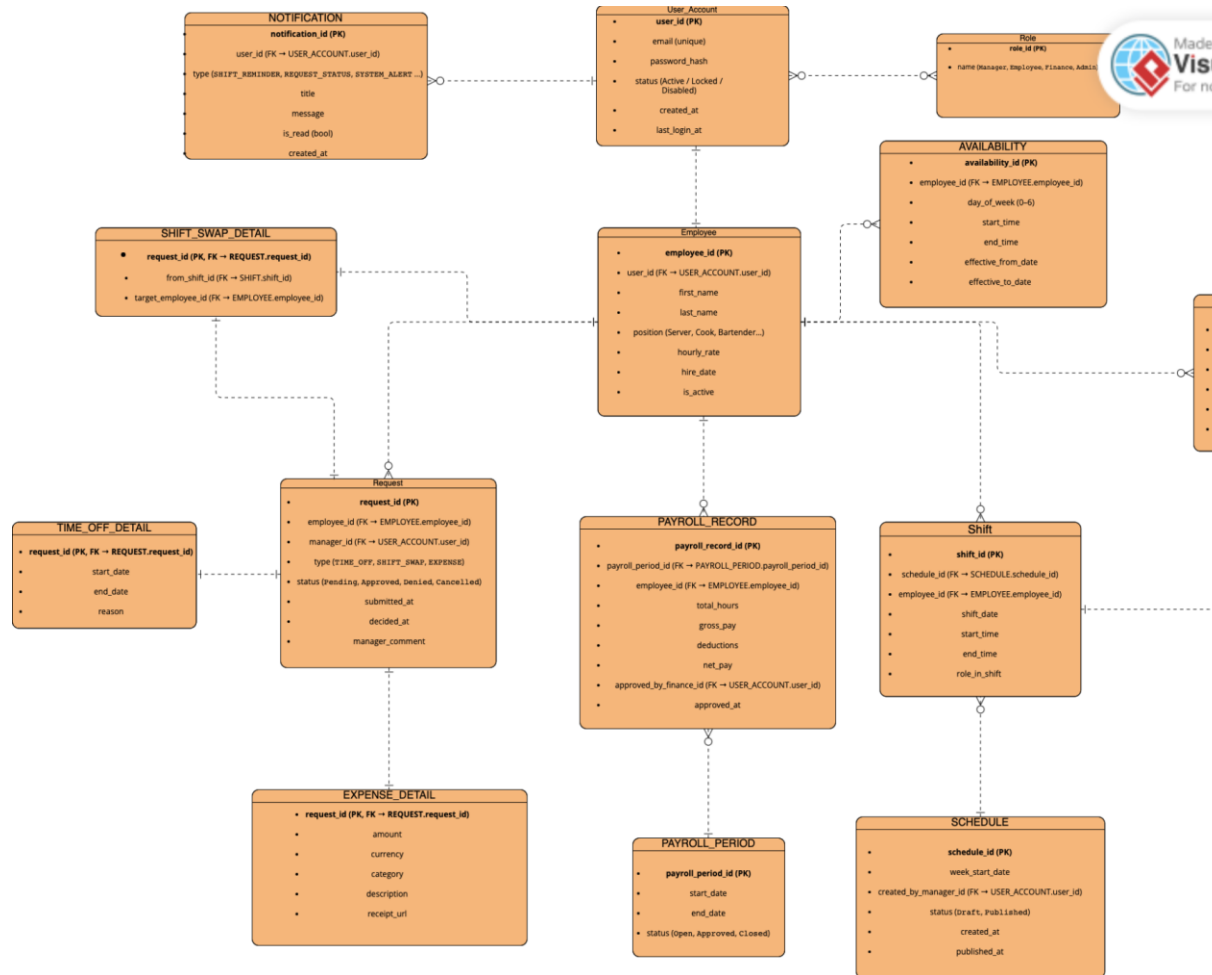
3.3 Data Modelling and Analysis

- Activity Diagrams

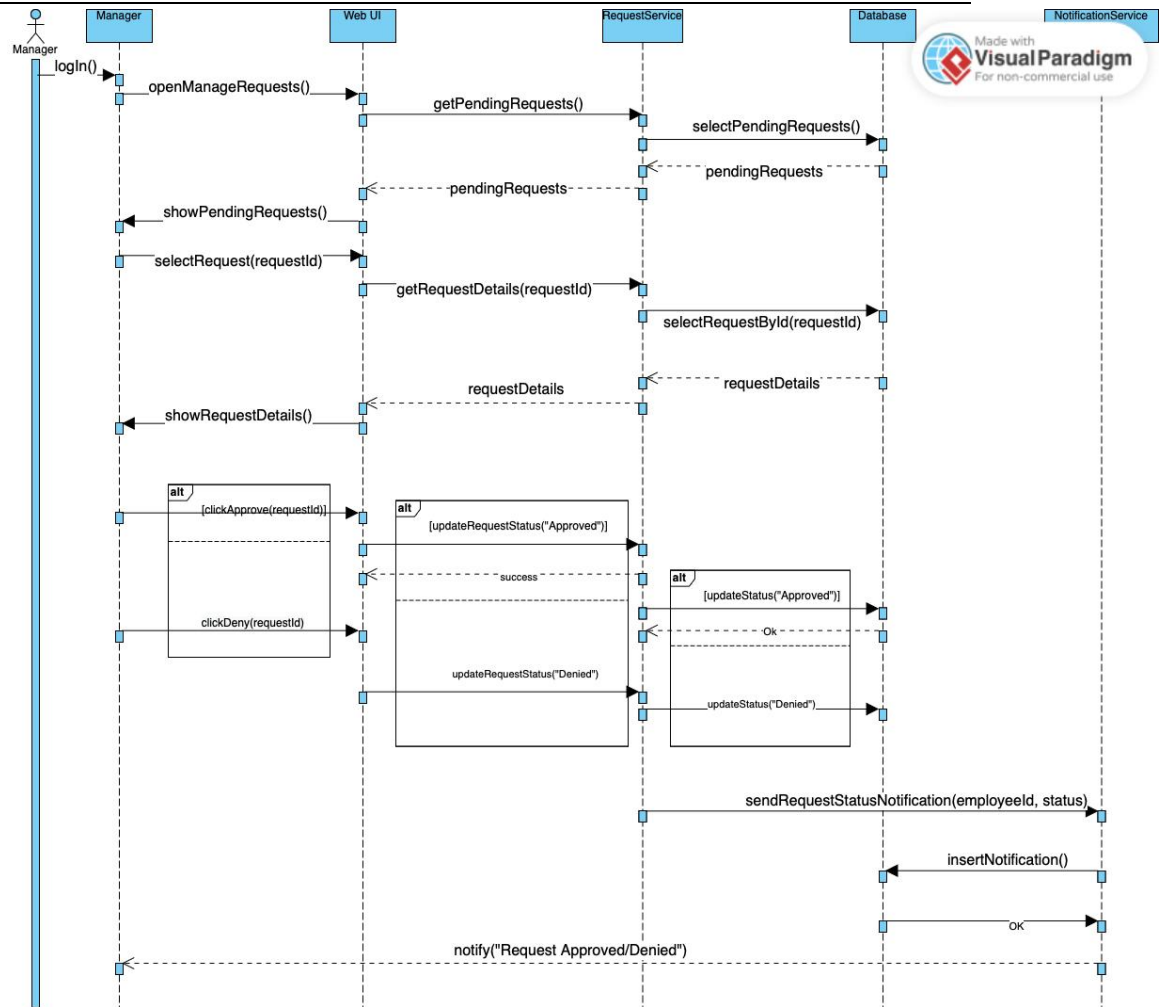




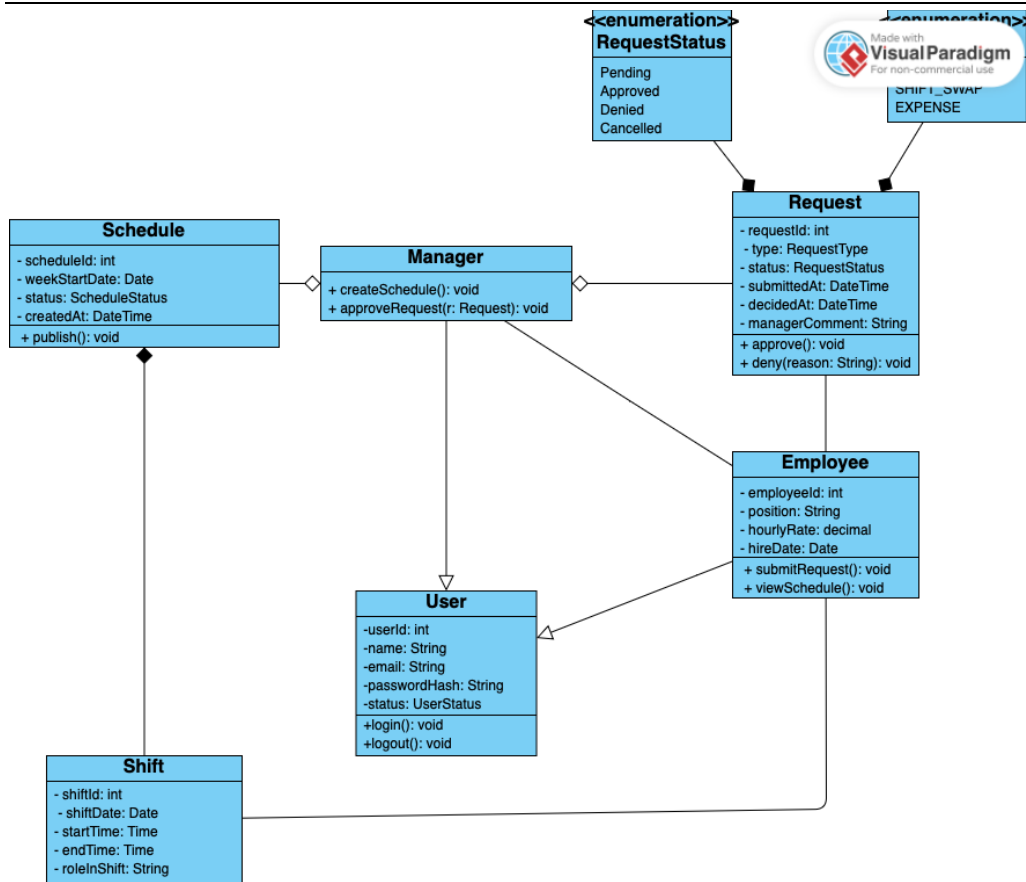
- Normalized Data Model Diagram



- Sequence Diagrams

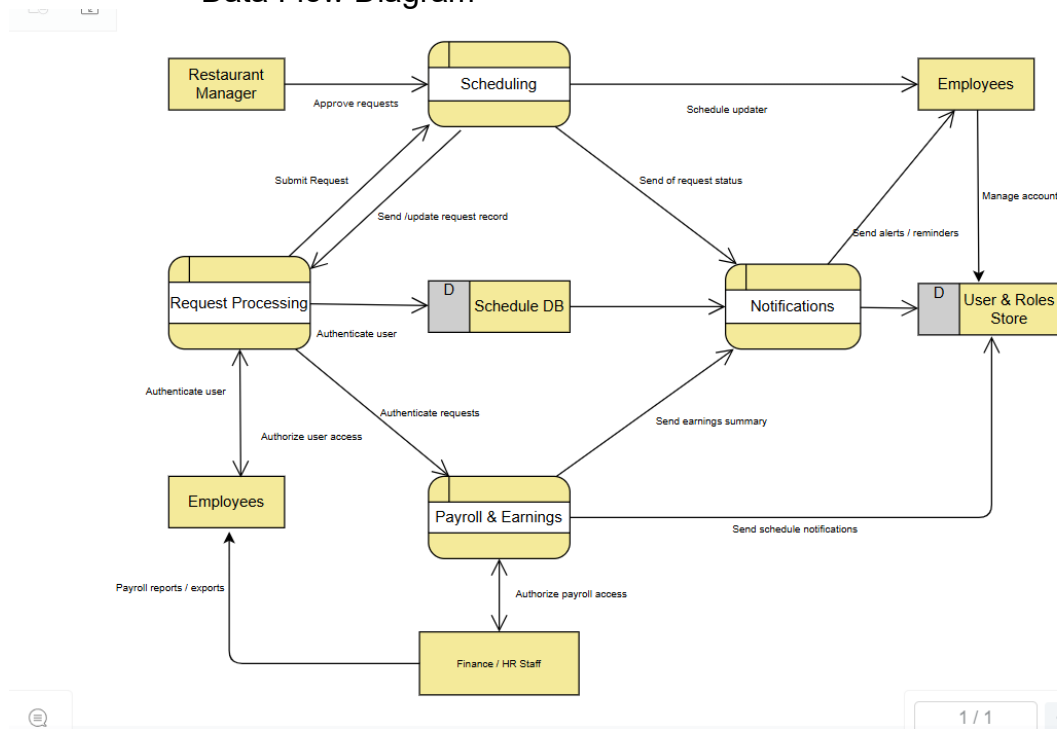


- UML Class Diagram



3.4 Process Modelling

• Data Flow Diagram



4.0 Non-Functional Requirements

The non-functional requirements for a system are typically constraints on the functional requirements – that is, not what the system does, but how it does it (e.g. how quickly, how efficiently, how easily from the user's perspective, etc.).

Non-functional requirements may exist for any of the following attributes – Performance, Reliability, Availability, Security, Maintainability, Portability.

Often these requirements must be achieved at a system-wide level rather than at a unit level. State the requirements in the following sections in measurable terms (e.g., 95% of transactions shall be processed in less than a second, system downtime may not exceed 1 minute per day, etc).

5.0 Logical Database Requirements

- Database Type: Relational (PostgreSQL)
- Data Integrity: All FKs enforced; cascade updates disabled for critical tables.
- Data Retention: Payroll records → 7 years; schedules → 2 years; logs → 6 months.
- Backup Policy: Nightly incremental + weekly full backup.
- Access Control: Roles = Admin, Manager, Employee, Finance with least-privilege permissions.

6.0 Other Requirements

- Scalability: Must support up to 100 active users simultaneously.
- Audit Logging: Every action stored with timestamp and user ID
- Error Handling: Graceful failures with user-friendly messages and error logs.
- Internationalization: English (default), ready for French translation.
- Deployment: Dockerized for portable setup on any server.

7.0 Approval

The signatures below indicate their approval of the contents of this document.

Project Role	Name	Signature	Date
PM & Tech. lead	Charles Langmuir	CL	2025-11-10
Software Development	Joel Chandra Paul	JCP	2025-11-10
Business Analyst	Yueyang Peng	YP	2025-11-10
QA & Testing lead	Jingyu He	JH	2025-11-10